KEYSIGHT

ADS Python API of RFPro Batch Pre-designed layout cells

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Agenda

1. Read me first
2. Implementation flowchart

exsting_layout_rfpro_py.zip

Read_Cell_Name_wrk.7zads

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Assumptions and Preconditions

- "The Python scripts are intended to be executed in Visual Studio Code, with the interpreter set to the python.exe found in the ADS installation path.“

- "In this example, existing layout cell names are distinguished and collected within libraries, and rfproviews are created for each library's cells, followed by sequential RFPro EM simulations.

- "This example assumes that the layout is designed based on a 2-port configuration, where p1 and p2 are the input and output ports, and p3 and p4 are the corresponding ground ports.
  It also considers a scenario in which multiple circuit cells and layout cells are mixed across within multiple libraries.“

- " Existing Layout means the user designed layout cells into workspace of ADS already, and the user want to run EM simulation automatically like a batch simulation with layout cells.”

- "If the three Python files mentioned above are handled by a single wrapper, errors may occur due to timing issues between each process."

Read me first

How to run this example

- VS code is used to run py files using python scripts
- To run this example, follo the below steps

1. Run 1_get_lib_cell_name.py *to collect library and layout cell names*

   Set the ADS directory, Set the HOME directory, Set the workspace name, then Run

2. Run 2_Top_rfpro_from_json_2.py *to generate rfproview, EM setup and to run simulation*

   Set the ADS directory, Set the HOME directory, Set the workspace name, then Run

3. Run 3_plot.py *to generate pdf for the RFPro results*

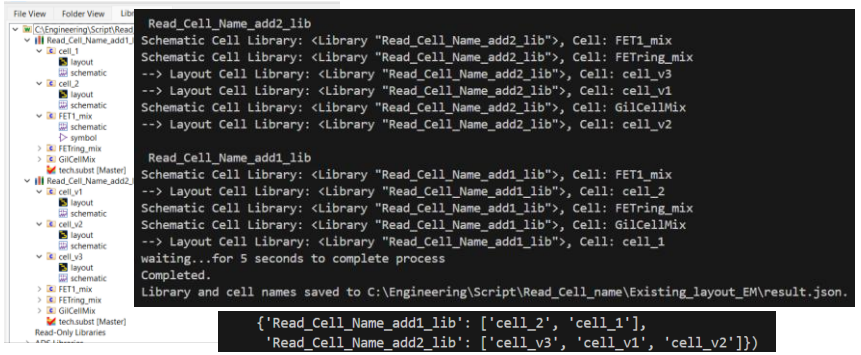
   Set the HOME directory, Set the workspace name, then Run

Flow chart of python files

1_get_lib_and_cell_name.py → result.json

: To collect library and Layout cell names from the workspace

: give names of layout cells and libraries



2_Top_rfpro_que_from_json_2.py → wrk_open.py

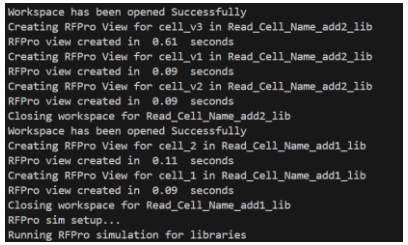
: To open workspace with information of lib and cell names

2_Top_rfpro_que_from_json_2.py → create_rfpro_view.py

: To create rfproview for each of layout cells

2_Top_rfpro_que_from_json_2.py → RFPro_EM_Setup_ads2026.py

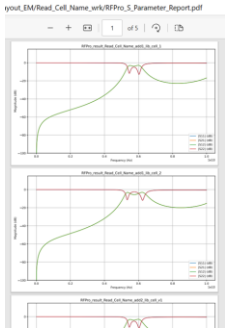
: To set EM setup for each rfproview



: rfpro simulation results

3_plot.py → RFPro_S_Parameter_Report.pdf

: PDF creation



Summary

- This example can be applied when a customer has pre-created layout cells for various reasons.To run this example, follo the below steps
- Python script, it is possible to identify the names of layout cells among the available cells and automatically generate an EM simulation batch.
- Batch generation is also possible even when multiple libraries and multiple cells are mixed.
- A possible next task could involve generating batches based on data, depending on how cells differ in port count and configuration.